



ROBOTS MEET THE ARTS

EDUCATIONAL ROBOTICS AND CODING MEET THE ARTS AND HUMANITIES IN INCLUSIVE LEARNING ENVIRONMENTS

MODULE 1

INTRODUCTION TO EDUCATIONAL ROBOTICS AND CODING

COORDINATOR AND PARTNERS LAB

STIMMULI
for social change



Co-funded by
the European Union

Co-financed by the European Union (project number: 2023-1-TR01-KA220-SC1-000151881). However, the opinions and views expressed are solely those of the author(s) and do not necessarily reflect those of the European Union or the European Education and Culture Executive Agency (EACEA). Neither the European Union nor the granting authority can be held responsible for them.



Theory and key concepts

Introduction

Computational thinking is the ability to think and solve problems in a structured way, following a logic similar to that of a computer. It consists of breaking down complex problems into smaller parts, identifying patterns, forming general rules and designing step-by-step instructions to find effective solutions. This process involves analyzing the situation, generating possible strategies and evaluating their results, thus promoting logical and creative reasoning.

The development of computational thinking from an early age promotes spatiotemporal orientation, abstract thinking and creative skills, stimulating imagination and problem solving.

This module introduces the concept of Computational Thinking and its main characteristics. Section 1: explores its educational potential and analyzes how to promote it in the classroom. Section 2: presents examples of disconnected and connected activities to work on it in a practical way.

Finally, an overview of educational robots and the most suitable programming environments for each educational stage is offered.

Introduction to Computational Thinking

Computational thinking is a way of solving problems like a computer scientist. It means breaking down large problems into smaller parts, finding patterns, establishing general rules, and creating step-by-step instructions for solving them. This helps you think logically and find creative solutions.

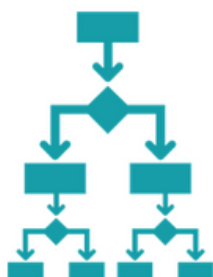
The concept of computational thinking was popularized by computer scientist Jeanette Wing in 2006, who emphasized its importance not only for computer scientists, but for everyone.



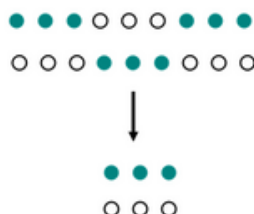
Since then, computational thinking has become an essential skill in education and many professional sectors, helping people to tackle challenges in a structured, clear and efficient way. This way of thinking is useful in many aspects of daily life, from learning and work to solving everyday problems.

What are the characteristics of computational thinking?

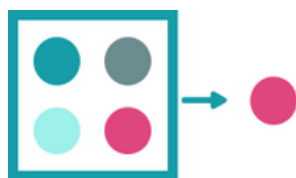
There are four key skills in computational thinking. They are:



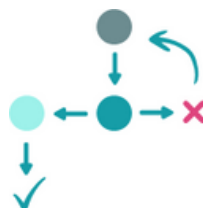
Decomposition
Divide a complex problem into smaller, simpler parts.



Pattern recognition
Find similarities between different problems or within the same problem.



Abstraction
Focus only on the important information, ignoring irrelevant details.



Algorithms
Develop a step-by-step solution or instructions to follow to solve a problem.



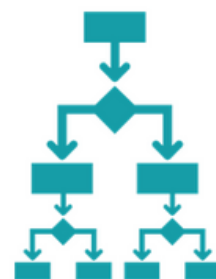
1. Decomposition

Procedure by which a complex problem is broken down into smaller parts, which are easier to solve.

Example:

Packing for a trip to Iceland:

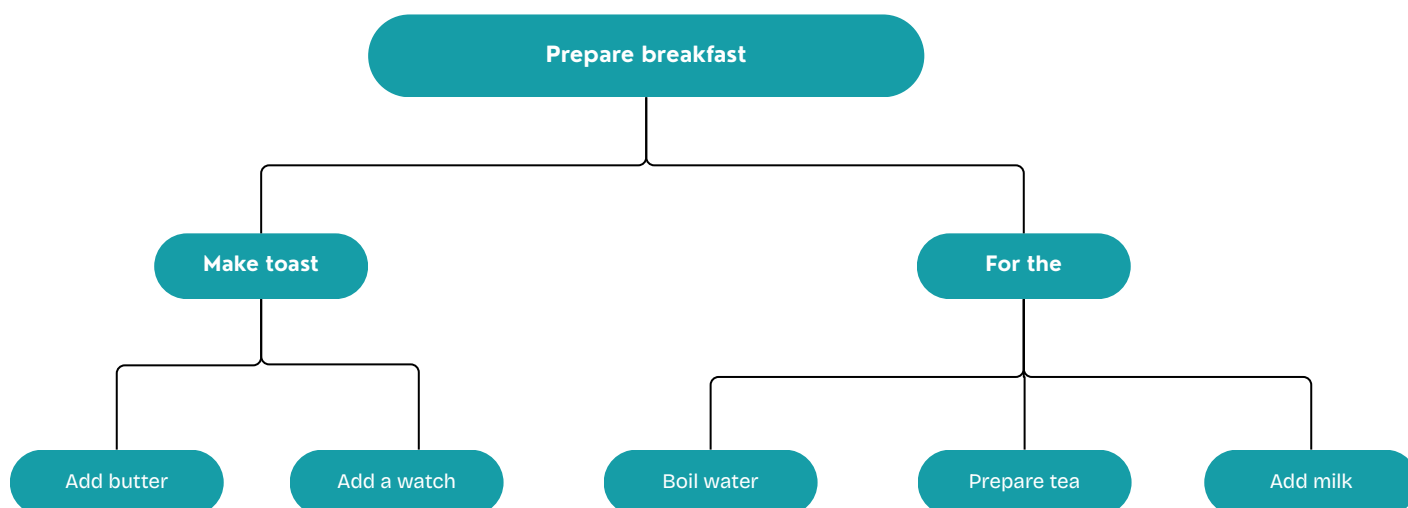
- Clothing: sweaters, pants, underwear, pajamas, jacket, scarf, gloves and hat.
- Footwear: hiking boots, comfortable shoes and sneakers.
- Toiletry kit: comb or brush, shampoo, soap, toothpaste and toothbrush, moisturizer and personal products.
- Documentation: passport or ID card, hotel reservations, boarding pass, travel insurance and bank cards.



Decomposition is a fundamental skill that allows you to tackle complex tasks more efficiently. By learning to divide a large problem into smaller, more manageable parts, students develop the ability to plan, organize, and delegate tasks within group projects, while improving time management. This approach not only facilitates problem solving, but also fosters confidence, autonomy, and efficiency when facing new challenges.

ACTIVITY

Use a decision tree to show how you break down an action, such as making breakfast.



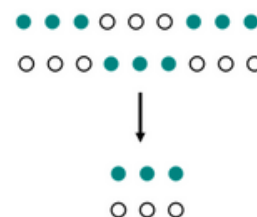


2. Pattern recognition

Identifying patterns that can be solved in a similar way to help solve complex problems.

Example:

Recognition of geometric patterns, numerical patterns and general rules.



Geometric pattern	Numeric pattern	General rule
  	1, 2, 3,...	$f(n) = n$
  	2, 4, 6,...	$f(n) = 2n$
  	4, 7, 10,...	$f(n) = 1 + 3n$
  	5, 9, 13,...	$f(n) = 1 + 4n$
  	8, 10, 12,...(verde) 9, 12, 15,...(total)	$f(n) = 5 + 3n$ $f(n) = 6 + 3n$

When a problem can be divided into several smaller parts, we often discover that some of these share common elements or characteristics. Even when they are not identical, they may resemble parts of other problems that we have already solved. Identifying these similarities makes it easier to understand and manage each component. Pattern recognition consists precisely of detecting these regularities and taking advantage of previous experience to address new situations more efficiently. This skill is very useful for making decisions and managing different contexts or challenges more easily.

For example, if you have ever learned a new language, you know how important patterns are. At first, understanding and memorizing vocabulary or grammar rules can be slow and complex. However, as you progress, you begin to identify recurring patterns, such as similar sentence structures or verb conjugations. This recognition allows you to learn new vocabulary more easily and apply grammar rules more accurately in different contexts. Ultimately, the initial experience provides you with tools and patterns that, over time, increase your efficiency and your ability to solve problems.





2. Pattern recognition

“

This pattern recognition skills are applicable **across various subjects** enhancing learning and problem-solving abilities.



Here's how pattern recognition is used in different areas of study:

English Language Arts

Students identify poems by recognizing similarities in examples, helping them understand poetic structures and themes.

Social Studies

Students analyze economic trends by examining data, allowing them to predict potential impacts and understand historical patterns.

Music

Students recognize patterns in musical compositions, such as recurring motifs or chord progressions, which helps in music theory and performance.

Geography

Students observe patterns in climate data to understand weather trends and predict future changes, aiding in environmental studies and planning.

Mathematics

Students use specific formulas to calculate slopes and intercepts, making it easier to solve complex equations and understand graphing.

Languages

Students group words by roots to better understand vocabulary, improving their ability to learn and use new words.

Physical Education:

Students identify common techniques in sports to improve their skills, like similar movements in different types of throws or kicks.

Science

Students classify animals by their characteristics and define common traits, aiding in the understanding of biological taxonomy and evolution.

Arts

Students categorize paintings by common artistic aesthetics and key characteristics, enhancing their appreciation and critique of different art styles.

Technology

Students detect recurring issues in coding errors or hardware problems, improving their troubleshooting and problem-solving skills.

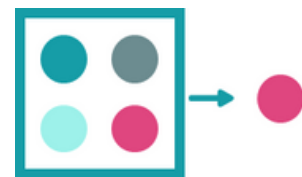


By integrating pattern recognition into their studies, students develop a deeper understanding of their subjects and enhance their ability to apply knowledge in practical situations.



3. Abstraction

Identify essential information and discard that which is not relevant in order to solve a problem efficiently.



Example:

Identify, among the many pieces available, those that are essential to build a figure.



Classify geometric figures of different colors. They can be grouped according to their shape, regardless of color, or according to color, without taking into account shape.

The activity can also be carried out with natural materials, such as leaves, shells, snails or stones.

Real-life abstraction example: using a washing machine

Using a washing machine is a good example of abstraction in everyday life. When doing laundry, you only need to follow a few simple steps: sort the clothes, load them into the washing machine, add detergent, choose the wash program, and press the start button. You don't need to understand the mechanical or electronic processes that take place inside the machine.

Abstraction simplifies all the complex functions—like filling with water, heating, spinning, and rinsing—into a few basic actions. This way, you can wash your clothes easily and efficiently without having to worry about the inner workings of your washing machine.



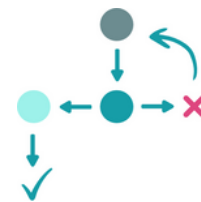


4. Algorithms

It defines a series of instructions that, followed in a specific order, allow a problem to be solved or a process to be automated.

Example:

Follow the instructions to perform a choreography.



Algorithm design

When a solution to a problem is available, it can be expressed in clear steps and executed sequentially, ensuring that the expected results are obtained consistently. Designing an algorithm consists of defining these steps and the rules that must be followed to achieve the same result every time.

Example: Cooking a recipe

Following a recipe is a clear example of algorithmic design. When you prepare a dish, you execute a series of defined steps to achieve the desired result. Each step is planned to ensure that the dish comes out the same every time you cook it.



Example: Step by step per fer creps

Gather the ingredients: 1 cup flour, 2 tablespoons sugar, 2 teaspoons baking powder, ½ teaspoon salt, 1 cup milk, 1 egg, 2 tablespoons melted butter.

Mix the dry ingredients: In a bowl, combine flour, sugar, yeast, and salt.

Mix the wet ingredients: In another bowl, whisk together the milk, egg, and melted butter.

Combine the mixtures: Pour the wet ingredients into the dry ones and stir until well combined.

Heat the pan: Preheat a non-stick pan over medium heat.

Cook the pancakes: Pour ¼ cup of batter per pancake. Cook until bubbles appear on the surface, then flip and cook until golden brown.

Serve: Serve the pancakes hot with your favorite accompaniments.

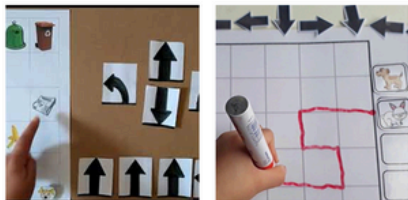


By following these steps, you ensure consistent results every time, demonstrating how algorithm design allows for predictable results. This same approach can be applied to many everyday tasks, from cooking to planning complex projects.



How can computational thinking be developed?

Unplugged Activities



Plugged Activities



Programming environments



Robots

What is its educational potential?

Creativity

Logical
Reasoning

Critical
Thinking

TeamWork

Oral
Communication

Soft Skills

What is educational robotics?

"Discipline that aims to generate heuristic learning environments based fundamentally on the active participation of students, generating learning from their own experience during the process of constructing and robotizing objects"

- Knowledge of programming languages
- Development of computational thinking
- Development of scientific skills
- Positive attitude towards technology
- Innovation and fostering creativity
- Inclusion



González, S.M. (2011). Study on the usefulness of educational robotics from the teacher's perspective. *Revista de Pedagogia*, 32(90), 81-117.

[Link](#)

How to use educational robotics in the classroom?

Learning robotics

Students use robots as a platform to learn robotics from a technical and engineering perspective.

Learning with robotics

Robotics and learning processes are used as tools to assist or facilitate teaching.

Learning through robotics

Students develop cross-curricular skills and work on content from different disciplines through robotics.

This latter paradigm is called robotics-based learning, in which the robot is an active tool that unifies the different dimensions of the educational process.



Gaudiello, I., and Zibetti, E. (2016). Learning robotics, with robotics, through robotics. *Learning robotics, with robotics, through robotics: educational robotics* (Vol. 3). Wiley Online Library. <https://doi.org/10.1002/9781119335740>



What is educational robotics for?

- Educational robotics is a tool that facilitates interdisciplinary work, such as in STEM fields.
- Its use focuses on developing 21st century skills, promoting mathematical, analytical and design skills, among others.
- With students at higher levels, it also allows for the exploration of advanced technologies.
- In addition, it incorporates game and gamification elements, offering students the opportunity to learn in a practical and fun way.

Benefits of educational robotics

- Teamwork: Promotes collaboration and socialization among students.

Confidence: Facing more complex tasks helps develop self-confidence.

Entrepreneurship: Experimentation and trial and error foster creativity, innovation and entrepreneurial skills in one's own projects.

Logical thinking: Robotics stimulates logical reasoning and structured problem solving.

Disadvantages of educational robotics

- Economical: Some robots can be expensive for schools, both in terms of acquisition and maintenance.
- Training: The lack of teacher training is a key factor for its correct implementation in the classroom.

Paradigm shift: The introduction of this technology requires a paradigm shift in teaching methodologies.

Appropriate use: Inappropriate implementation of robotics can limit access and generate inequalities in learning.



Robots for each educational stage



Bee-bot



Blue-bot



Lego WeDo



Edison



Lego Mindstorms

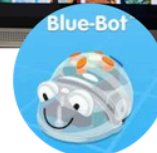
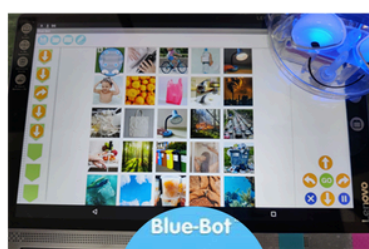
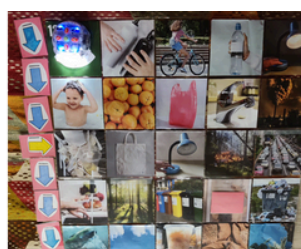


Dash and Dot

Ground robots for early childhood education



Blue-bot



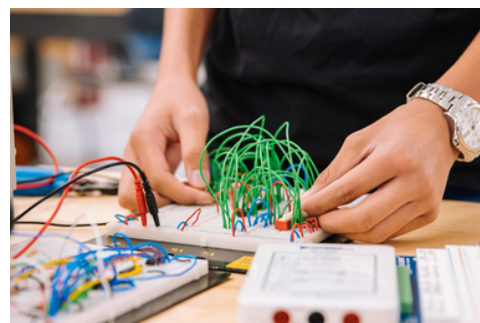


Ground robots for primary education

Once students have assimilated the basic concepts of programming and how robots work, they can begin to use more advanced robots and programming environments, applying them to more cross-curricular content.

- In the first years of primary education, it is advisable to use robots that can be programmed using a computer or a tablet application.

In the last stage of primary school, boards such as Arduino, Lego Mindstorms or others similar can be introduced.



Robots according to educational stage

Technological resources	STAGE (years)						
	EI (3-5)	CI (6-7)	CM (8-9)	CS (10-11)	ESO (12-13)	ESO (14-15)	BAT / CF (+ 16)
Bee-Bot	x	x					
Blue-Bot	x	x					
Code Mouse	x	x					
Dash&Dot		x	x	x			
Lego Wedo		x	x				
Edison			x	x			
Lego Mindstorm					x	x	
3D bot Scratch				x	x		
3D bot Arduino					x	x	x
Robomind						x	x

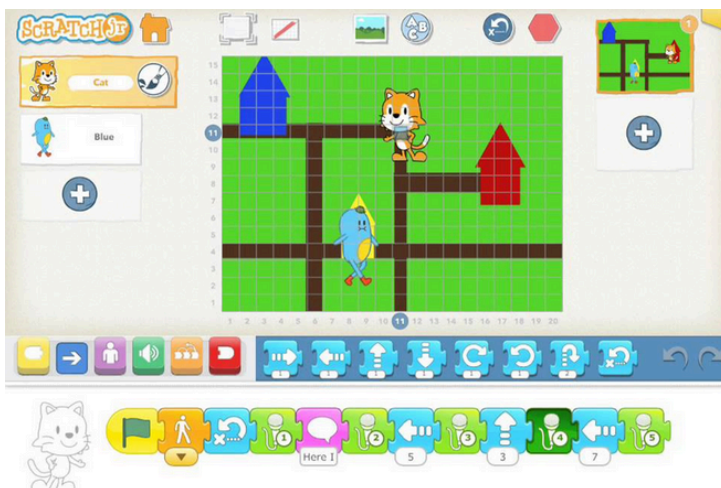
Programming environments

Once students have assimilated the basic concepts of programming and how robots work, they can use more advanced robots and programming environments, applying them to more integrated and cross-curricular content.



<https://www.scratchjr.org/>

5-7 YEARS Available for tablet or mobile



8 - 16 YEARS Available by computer

<https://scratch.mit.edu>

